Department of Computer Science and Information Technology

CT-499 Final Year Design Project

Proposal for the Final Year Design Project

Title	Salesforce Mobile Business Analytics & Forecasting Application

Domain	Domain 1	Domain 2	Domain 3	Domain 4	Domain 5	Domain 6
	Data Science	Business Analytics	Customer Relationship Management (CRM)	Cloud Computing	Mobile Application Development	Business Intelligence

1. Nature of Project [Tick all that applicable]

☑New Project OR ☐ Extension of Existing Project	☑Industrial Collaboration	☐ Funded
☐ Other Department Collaboration (If yes) Department Name_____	☐ Other Academic Institution Collaboration (If yes) Institution Name_____	

2. Brief Outline (*Problem Identification and Significance*)

- Businesses often face limited access to business data on mobile devices, difficulty monitoring sales performance, inefficient tracking of leads and opportunities, and delayed decision-making due to outdated or insufficient reports.
- Existing reporting processes may not provide sufficiently accurate and interconnected data for effective analytics, limiting the ability of business users to obtain timely insights.
- This project addresses these challenges through a Salesforce-based mobile business analytics and forecasting application that integrates lead management, opportunity tracking, KPI monitoring, interactive dashboards, notifications, and mobile decision support within the Salesforce ecosystem.
- The project will use a general/demo business scenario designed to provide a demonstrable solution that can potentially be presented to different clients rather than representing a confidential client-specific implementation.

3. Objectives

- Enable business users to access relevant Salesforce business data and analytics through a mobile-oriented interface.
- Provide lead management and lead-conversion insights for monitoring sales prospects.
- Implement opportunity tracking including stages, amount, expected closing dates, and pipeline performance.
- Develop interactive KPI dashboards for sales, revenue, leads, opportunities, conversion, target achievement, and growth.
- Automate important business events and notifications using Salesforce Flows.
- Provide role-specific dashboards, data visibility, permissions, and security for executives, sales managers etc.
- Support faster data-driven decision-making by providing timely business insights through the Salesforce mobile ecosystem.

4. Scope

The application includes:

Mobile Business Application

- Salesforce mobile access with a mobile-optimized user experience.
- Mobile access to relevant dashboards, leads, opportunities, and business insights.

Lead Management

- Access and monitor lead information.
- Support lead-related monitoring and follow-up requirements within the Salesforce environment.
- Provide visibility of lead volume and lead conversion performance were supported by available data.

Opportunity Management

- Track opportunity stages and pipeline progress.
- Display opportunity amount and expected closing dates.

Business Analytics & KPI Monitoring

- Interactive dashboards for prioritized KPIs, including lead conversion/total leads, revenue, total sales.
- Filters and date-range selection.
- Salesperson and region-based filtering where applicable.
- Drill-down into detailed records.
- Interactive charts and visual business insights.

Notifications & Automation

- Automated notifications for new leads.
- Alerts for opportunity updates and deal-stage changes.
- Target missed/achieved notifications.
- Follow-up reminders.
- Opportunity closing-date alerts.
- Automation implemented using Salesforce Flows.

Role-Based Access & Security

- Role-specific data visibility and access permissions.
- Support access requirements for executives, sales managers, and sales representatives.
- Follow Salesforce controls for user permissions, record visibility, and protection of business data.

Salesforce Configuration & Development

- Use standard Salesforce objects including Lead, Opportunity, and Account.
- Create custom fields where required; no custom objects are required.
- Use Salesforce Reports, Dashboards, Lightning Pages, Flows, and other available platform functionality.
- Use Apex only where Salesforce configuration/features are insufficient for a requirement.

Data

- Use sample/demo data provided for the project by Salesforce.
  Use available historical sales, revenue, opportunity, and lead-conversion data for analytics.

5. Proposed Methodology

- Analyze the clarified business requirements, priorities, user roles, KPIs, and Salesforce capabilities.
- Review the available Salesforce partner organization and configure the required development environment.
- Prepare and validate sample/demo data and historical records required for analytics.
- Configure Salesforce Lead, Opportunity, and Account data structures and create required custom fields.
- Design mobile-friendly Lightning pages and Salesforce mobile views for target users.
- Develop Reports and Dashboards for KPI monitoring, interactive analytics, filtering, drill-downs, and performance visualization.
- Configure opportunity monitoring for stage, amount, expected closing date.
- Configure Salesforce Flows to automate notifications, reminders, and specified business events.
- Apply role-based access, permissions, and record visibility according to user requirements.
- Use Apex or Salesforce Einstein only where required after evaluating native Salesforce functionality; external APIs/third-party applications remain fallback options.
- Conduct functional, usability, mobile, security/access, dashboard, automation using sample/demo data.
- Demonstrate the integrated Salesforce mobile analytics.

6. Resources Involved

Salesforce Platform:

- Salesforce Partner Organization.
- Salesforce Mobile App.
- Lightning App Builder / Lightning Pages.
- Salesforce Reports.
- Salesforce Dashboards.
- Salesforce Flows.
- Standard Salesforce Objects.
- Custom Salesforce Fields.

Development & Analytics:

- Apex
- Salesforce APIs, only where required as a fallback.

Data:

- Sample/demo business data.
- Historical sales data.
- Revenue records.
- Opportunity history.
- Lead-conversion history.

7. Description of Industrial Support (If any)

- Nespon Solution will provide a Salesforce partner organization for development.
- The FYP team will have access to relevant Salesforce capabilities, including configuration of objects/fields, Reports and Dashboards, Flows, Lightning Pages, and Apex development.
- Nespon Solution will provide requirement clarification and guidance regarding project priorities and expected demonstration outcomes.
- Sample/demo data and historical data required for analytics will be made available for the project.

8. SDGs (If Applicable)

☐ No Poverty	☐ Zero Hunger
☐ Good Health and Well-Being	☐ Quality Education
☐ Gender Equality	☐ Clean water and Sanitation
☐ Affordable and Clean Energy	☑Decent Work and Economic growth
☑Industry, Innovations and Infrastructure	☐ Reduced Inequalities
☐ Sustainable Cities and Communities	☐ Responsible Consumption and Production
☐ Climate action	☐ Life Below Water
☐ Life on Land	☐ Peace, Justice and Strong Institutions
☐ Partnerships	

9. Gantt Chart

Year	2026					2027				
Semester	Phase 1					Phase 2				
Months	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Literature Review & Requirement Analysis										
Proposal Preparation, Submission & Approval										
Requirement Gathering										
System Design & Salesforce Architecture										
Salesforce Environment & Data Preparation										
Security, Roles & Access Control										
Mobile UI Mockup										
Business Analytics, KPIs & Dashboards										
Lead & Opportunity Management										
Notifications & Workflow Automation										
Front end										
System Integration & Functional Testing										
Mobile Usability & Performance Testing										
Documentation, Report Writing & Final Refinement										
Final Demonstration & Evaluation										

10. Details of Project Team

i. Students

No.	Name	Seat No.	Signature (s)
1	Wajiha Adnan	DT-23001	
2	Muhammad Abdullah Kidwai	DT-23302	
3	Taaha Nadeem	DT-23303	
4	Fizza Ahmed	DT-23304	

ii. Supervisors / Advisors

	Name	Designation & Department	Address & Contact	Signature(s)
Supervisor	Dr. Muhammad Imran	Assistant Professor	hmimran@cloud.neduet.edu.pk	
Co-Supervisor (If any)	Muhammad Hammad	Lecturer	hammad@cloud.neduet.edu.pk	
Industrial Advisor (If any)	Kauser Laiq	Developer at Nespon Solutions	kauser.laiq@nespon.com	Kauser Laiq

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Project Serial No.: _____ Dated: _____	Signature Convener Steering Committee	Signature FYP Coordinator

☐ Proposal Approved	☐ Not Approved	☐ Returned for Clarification / Modification
Comments: (if any)		

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(Signature of Chairperson)

Date: _____